Yassine Abassi

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Summary

Computer Engineering Master's student spanning full-stack web development and embedded/robotics systems. Experienced shipping production features end-to-end — from RESTful APIs and CI/CD pipelines to multi-robot coordination.

Education

Polytechnique Montréal — Montreal, QC
Master's in Computer Engineering — *Aug 2026 – Present*
Bachelor's in Computer Engineering — *Aug 2022 – May 2026*

Experience

FIU Center for Translational Science — Port St. Lucie, FL
Research Software Developer — *May 2026 – July 2026*

- Translated researcher needs into pipeline requirements, building a **Python/RDKit** cheminformatics tool converting chemical names to SMILES and assembling branch combinations onto a molecular scaffold under strict stereochemistry rules.
- Generalized the pipeline to support arbitrary scaffold branches parsed directly from Excel inputs, validating outputs and automating the export of a library exceeding **10,000+** compounds.

Hydro-Québec — Montreal, QC
Software Developer — Capstone Project — *Jan 2026 – May 2026*

- Collaborated closely with cross-functional stakeholders to design a full-stack weather visualization platform using **Angular** and **ASP.NET Core**, featuring geospatial mapping via **Leaflet** and real-time updates via **SignalR**.
- Documented system architecture and APIs; implemented **GDAL** raster processing, designed **RESTful APIs** backed by an **SQLite** database via **EF Core**, and managed end-to-end deployment via **Docker** and automated **CI/CD** pipelines.

UpToTest — Tunis, Tunisia
Software QA Automation Intern — *May 2024 – Sep 2024*

- Architected a **Cypress/Cucumber** BDD test framework using a **POM** design pattern, executing rigorous regression cycles across **400+** end-to-end scenarios to achieve **89%** feature coverage.
- Managed defect tracking, performed cross-browser validation against pre-production environments, and delivered sprint test reporting and Mochawesome HTML metrics alongside an Agile Scrum team.

Projects

3D Interactive Portfolio — Live Demo

- Crafted a 3D portfolio in **React.js** and **Three.js** featuring an interactive room with custom camera rigging.
- Expanded the user experience with a Gemini-powered chatbot assistant with **Upstash Redis**-backed rate limiting.

SSH Terminal Portfolio — Live Demo

- Built an SSH-based interactive terminal portfolio in **Go** using the **Charm** ecosystem.
- Deployed on **Fly.io** with **Docker** containerization, persistent **SSH** host key management, and automated **CI/CD**.

WildGuard — AI Wildlife Monitoring System — GitHub

- Built an AI-powered ecological surveillance platform in **Python** for anti-poaching and endangered species protection, with real-time detection updates via **Socket.IO** and automated SMS alerting through **Twilio**.
- Implemented a **YOLOv8** detection model capable of identifying **50+** species, integrated with a **Flask** backend.

Multi-Robot Exploration System

- Designed a coordinated multi-robot exploration system in **Python** for **AgileX Limo** robots using **ROS2** and **Gazebo** for simulation, implementing custom nodes for inter-robot communication and task distribution.
- Containerized software modules using **Docker** for reproducible deployment across physical and simulated environments.

Embedded Obstacle Detection System

- Built an autonomous obstacle detection and search system with **low-level I/O** control and real-time decision logic in **C++** on the **ATmega164A** microcontroller.
- Implemented a reusable static library, finite state machines, interrupts, timers and **PWM** motor control.

Technical Skills

Languages: C, C++, C#, Java, Python, JavaScript, TypeScript, Go, VHDL
Web Frameworks: Angular, React.js, Node.js, NestJS, Express.js, WebSockets, SignalR, .NET, Three.js
Data & APIs: SQL, PostgreSQL, MongoDB, SQLite, Firebase, Retrofit, EF Core, GDAL, Upstash Redis
DevOps & Tools: Git, GitHub Actions, GitLab CI, Docker, Jira, Fly.io, Cypress
Embedded & Hardware: Zynq7000, ATmega164A, Pynq-Z1, Basys3 Artix-7, AgileX Limo

Additional Information

Languages: English, French (Fluent) — German (Intermediate).
Work Authorization: Authorized to work in the U.S. and Canada (dual citizen), no sponsorship required.